Low Level Design On

Marketing Planning Assistant Agent

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1	Bhavya Rathore	EN22CS301269
2	Bhoomika Sanodiya	EN22CS301271
3	Bhumika Chouhan	EN22CS301275
4	Devi Sanjaykumar Modi	EN23CS3T1008
5	Dollyna Sahu	EN22CS301354

Group Name: Group G2D3

Project Number: AAI-26

Industry Mentor Name: Mr. Ashwin Krishnan C.K.

University Mentor Name: Prof. Maya Yadav

Academic Year: 2025-2026

Table of Contents

1. Introduction.
    - 1.1. Scope of the document.
    - 1.2. Intended Audience
    - 1.3. System overview.
2. System Design.
    - 2.1. Application Design
    - 2.2. Process Flow.
    - 2.3. Information Flow.
    - 2.4. Components Design
    - 2.5. Key Design Considerations
    - 2.6. API Catalogue.
3. References

1. Introduction

This document presents Marketing Planning Assistant Agent is an Agentic AI system built to automate the process of creating a complete marketing campaign plan from a single high- level goal. Traditional marketing planning requires a team of specialists — a researcher, a strategist, a scheduler, and a report writer working over several days to produce a campaign document. This project replaces that manual process with five AI agents that execute sequentially and deliver a structured marketing plan in under five minutes.

The project is built using CrewAI as the multi-agent orchestration framework, Groq as the free- tier LLM provider running the llama-3.1-8b-instant model, Python 3.12 as the programming language, and Streamlit for the web interface.

1.1. Scope of the document

This document defines the detailed low-level design of all agent, task, tool, and utility classes, Sequential data flow from user input through five AI agents to final output ,User flow of the Streamlit web application ,Entity definitions and relationships between system components,Technology stack selection .

1.2. Intended Audience

This document is intended for:

- Developers implementing the system
- Academic evaluators and mentors
- Students studying Agentic AI
- Researchers exploring AI in Marketing Planning Assistant Agents

It helps the audience understand system design, workflows, and technical implementation details.

1.3. System overview

The system is a multi-agent AI-based Marketing Planning assistant that processes user- defined goals and generates structured marketing plans.

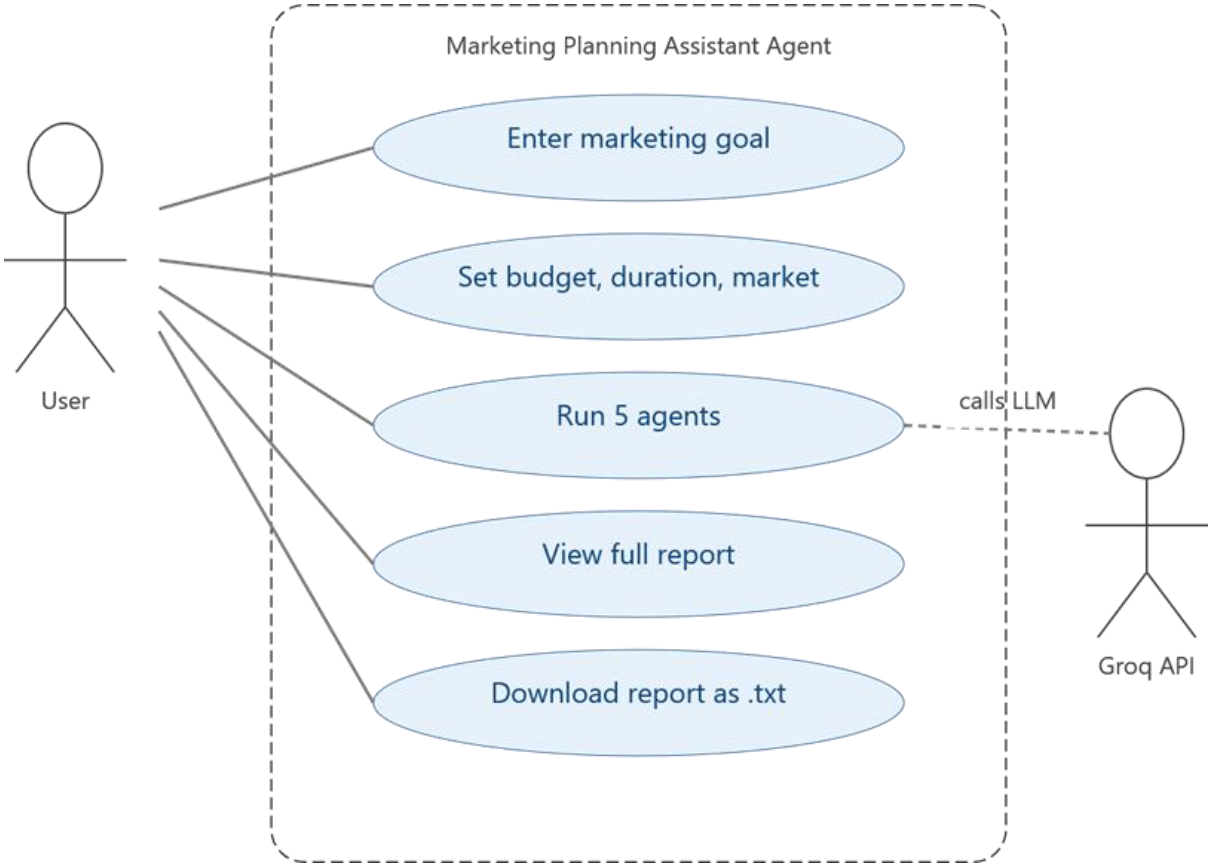

Figure 1: Use Case Diagram

Working Overview

1. User provides a Marketing goal
2. Planner Agent interprets the goal
3. Goal is decomposed into smaller tasks
4. Specialized agents perform tasks
5. Tools validate data and resources
6. Final execution plan and report is generated

Key Features

- Multi-agent collaboration

- Tool-based validation
- Dynamic planning and scheduling

2. System Design

The system follows a modular multi-agent architecture where each component performs a specific role. The design ensures flexibility, scalability, and efficient task execution.

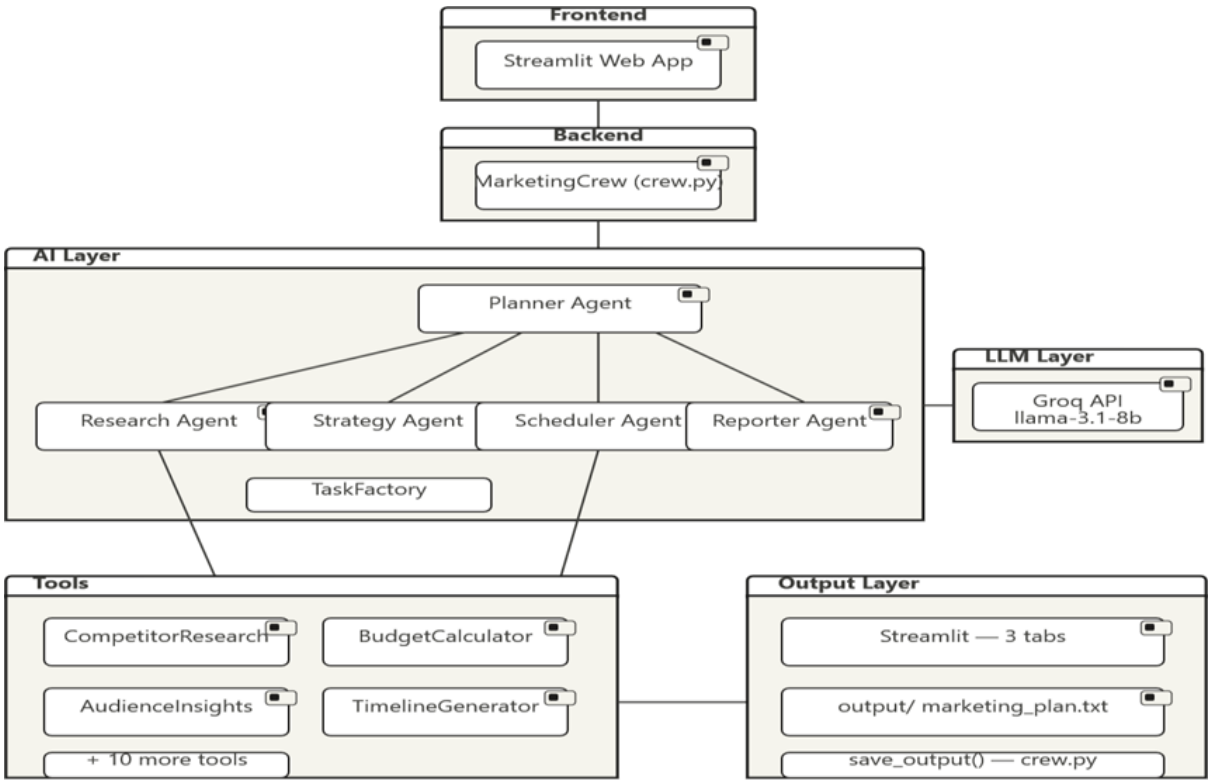

Figure 2: Architecture Diagram

Architecture Flow

User Input → Planner Agent → Task Decomposition → Agents → Tools → Output

2.1. Application Design

The application consists of the following modules:

1. **Input Module**

- Accepts marketing goal from user in natural language
- Validates that the goal is non-empty and meaningful
- Enriches input with additional parameters — budget, duration, and target market
- Passes the final goal string to the Planner Agent

2. **Planner Agent**

- Core decision-making unit
- Breaks goals into smaller tasks using LLM

3. **Agent Layer**

- Research Agent analyzes the market, identifies 3 competitors with their messaging and weaknesses, builds 2 audience personas, and finds 5 relevant market trends
- Strategy Agent builds the campaign strategy including positioning statement, channel mix, budget allocation across channels, and 7 measurable KPIs with targets and deadlines
- Scheduler Agent creates the 12-week execution timeline divided into 4 phases — Pre-Launch, Launch, Optimize, and Evaluate — with task owners and dependencies
- Reporter Agent aggregates all previous agent outputs and writes the complete 8- section executive marketing plan report

4. **Tool Layer**

- Competitor Research Tool
- Audience Insights Tool
- Market Trends Tool
- Budget Calculator Tool
- Channel Selector Tool

- KPI Generator Tool
- Timeline Generator Tool
- Dependency Mapper Tool
- Resource Allocator Tool
- Report Formatter Tool
- Risk Assessor Tool

5. Output Module

- Aggregates results
- Generates structured Marketing plan

2.2. Process Flow

The system follows a structured execution flow:

1. User inputs marketing goal
2. System validates input
3. Planner Agent decomposes the goal
4. Tasks are assigned to agents
5. Agents execute tasks
6. Tools are used for validation
7. Results are combined
8. Final plan is generated

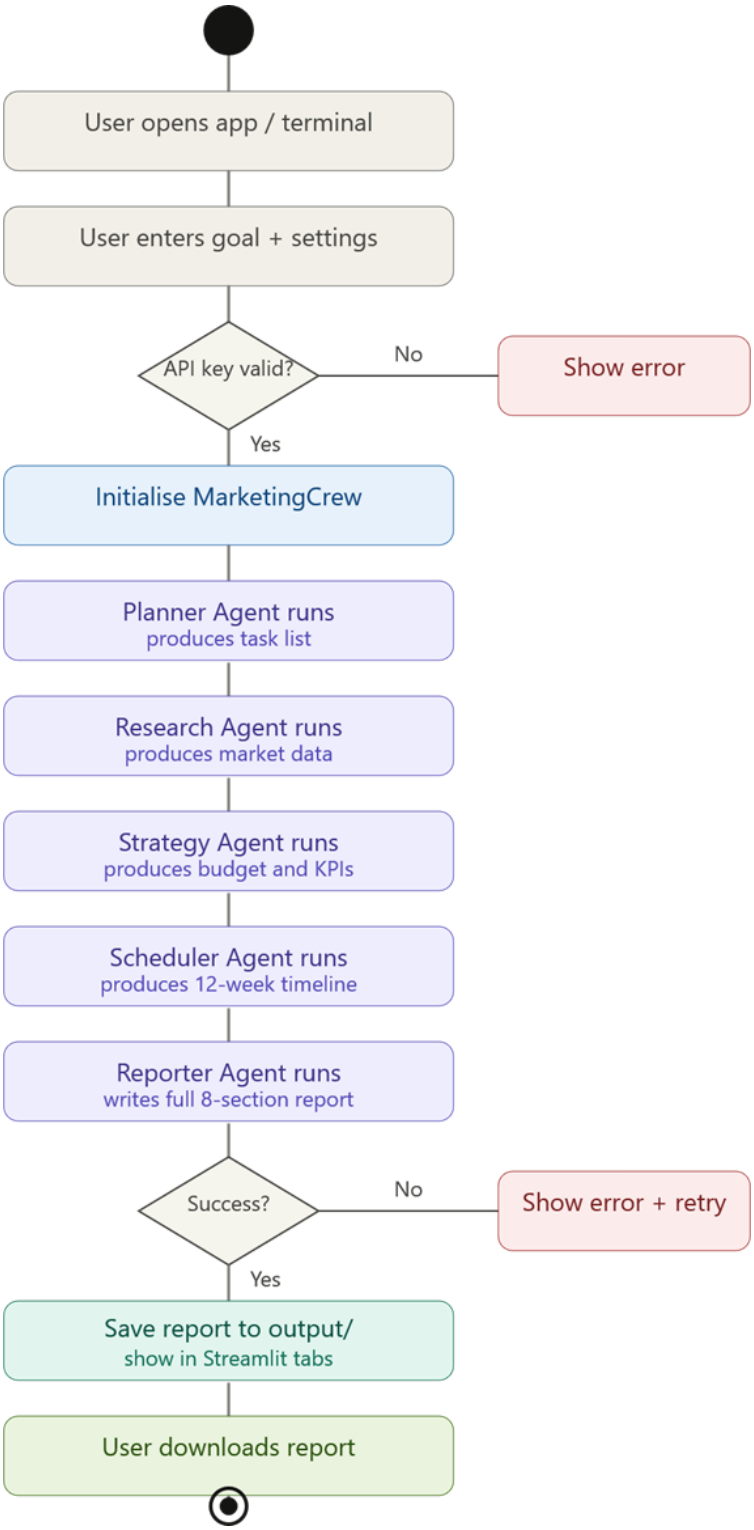

Figure 3: Activity Diagram

2.3 Information Flow

The system follows a sequential agent-based workflow, where data moves step-by-step from user input to final output. Each stage transforms the information, adding more context and refinement, resulting in a structured marketing plan. This flow ensures clarity, modular processing, and better control over each stage of execution.

Key Characteristics of Flow

- Sequential and controlled execution
- Context-aware (each step builds on previous)
- Modular and easy to extend
- Ensures structured and high-quality output The flow of information in the system is as follows:

User Input

↓

Planner Agent

↓

Task List

↓

Agents

↓

Tool Calls

↓

Processed Data

↓

Final Output

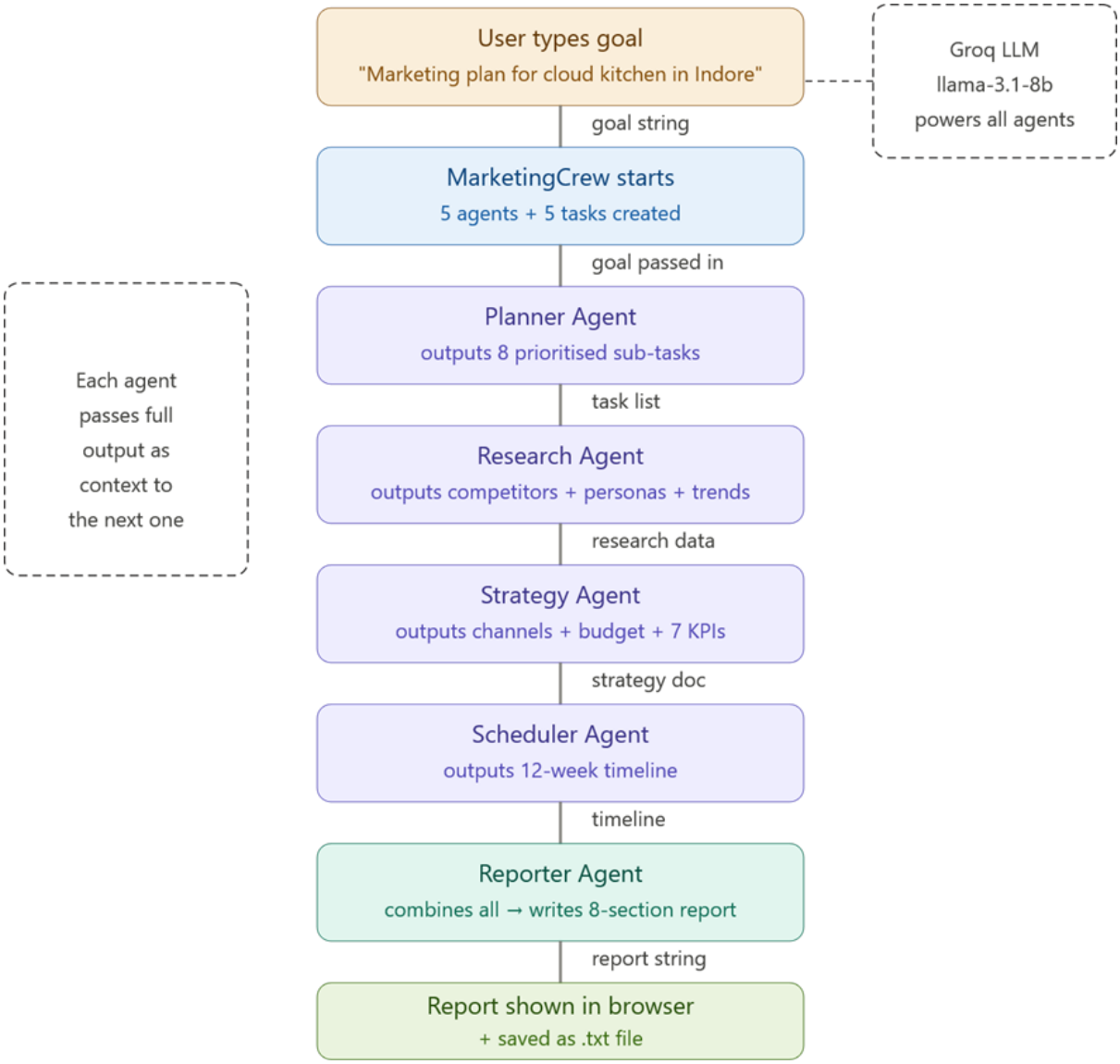

Figure 4: Flow Chart

Data Handling

- Input is processed and structured
- Intermediate data is shared between agents
- Final output is formatted for readability

2.4. Components Design

1. Planner Agent

- Responsible for task decomposition
- Uses LLM for reasoning and planning

2. Research agent

- Analyzes patient symptoms
- Identifies possible conditions

3. Strategy Agent

- Builds the full campaign strategy from research findings
- Selects the top 3 to 4 marketing channels with budget justification

4. Scheduler Agent

- Converts strategy into a 12-week execution timeline
- Divides work into 4 phases — Pre-Launch (Weeks 1–3), Launch (Weeks 4–6), Optimize (Weeks 7–10), Evaluate (Weeks 11–12)
- Assigns owner roles to each task and identifies critical path dependencies

5. Reporter Agent

- Synthesizes all previous agent outputs into a polished executive document
- Writes 8 sections — Executive Summary, Goal, Research, Strategy, Schedule, Risks, Quick Wins, Appendix
- Adds risk assessment with top 3 risks and mitigations

6. LLM Integration

- Uses Groq API
- Model: llama3-8b-8192
- Temperature = 0.3

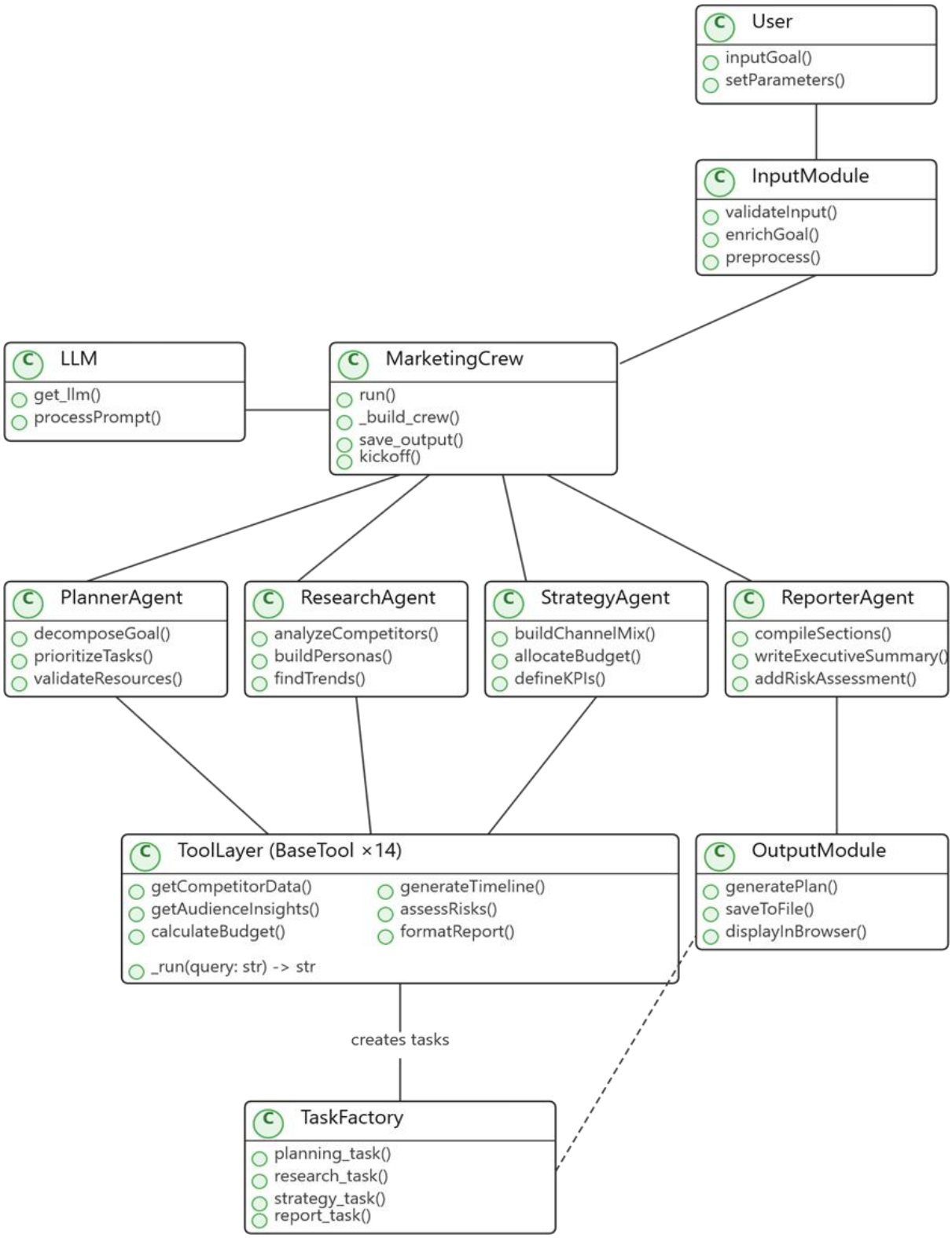

Figure 5: Class Diagram

2.5 Key Design Considerations

The system is designed with a focus on simplicity, scalability, and modular architecture. It ensures that components can evolve independently while maintaining overall system efficiency. These considerations help in making the system easy to extend, maintain, and deploy, especially when transitioning from prototype to production.

Scalability

- Supports multiple agents working in sequence
- Can be extended to handle parallel execution in production
- Scales with better backend (e.g., FastAPI + queue systems)

Modularity

- Each agent and component is independent
- Easy to update or replace individual modules
- Reduces impact of changes on the overall system

Flexibility

- New agents and tools can be added easily
- Supports different marketing use cases
- Configurable inputs and workflows

Performance

- Uses efficient LLM models for faster response
- Background execution keeps UI responsive
- Optimized for prototype-level speed

Reliability

- Mock tools reduce dependency on external APIs
- Ensures stable and predictable outputs
- Minimizes failure points during execution

Security

- No real or sensitive user data is used
- Local execution ensures data privacy
- Minimal external data exposure

2.6 API Catalogue

The system integrates both external and mock APIs to perform various tasks such as content generation, research, and planning. These APIs provide structured inputs and outputs, enabling smooth communication between system components.

1. LLM API (Groq)

Used for generating responses and executing agent tasks using a large language model.

Details:

- API Key → Used for authentication
- Model → llama3-8b-8192
- Input → Prompt (user goal + context)
- Output → Generated response text

2. Competitor Research API (Mock)

Simulates competitor analysis for a given industry or market.

Input:

- Industry or market keyword

Output:

- Competitor profiles including:
  - Ad spend
  - Marketing channels
  - Messaging strategy
  - Weaknesses

3. Audience Insight API (Mock)

Provides insights about the target audience based on the product or campaign.

Input:

- Product description or target market

Output:

- Audience insights such as:
  - Demographics
  - Preferences
  - Behavior patterns
  - Pain points

4. Budget Calculator API (Mock)

Helps in distributing the campaign budget across different channels.

Input:

- Campaign goal type
- Total budget

Output:

- Channel-wise budget allocation:
  - Percentage split
  - Suggested spending per channel
  - Basic reasoning for allocation

3. References

- CrewAI Documentation
- LangChain Documentation
- Groq API Documentation
- Research papers on Agentic AI
- Marketing Planning Assistant AI system design resources